Pearson BTEC

Mark Scheme (Results)

June 2024

Pearson BTEC Nationals
In Computing (31769H)
Unit 2: Fundamentals of Computer Systems

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Unit 2: Fundamentals of Computer Systems

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

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Question Number	Answer	Mark
1a	Any **three** from: • Can be more easily upgraded. (1) • (Usually) cheaper than the same specification laptop / less expensive to upgrade. (1) • Individual peripherals (e.g. keyboard, screen) can be replaced if broken (1) • Case can be hidden to avoid damage from/injury to young children. (1) • Screens are often larger • Less likely to be stolen/lost (because of size/fixed location) (1) • Usually have better/more effective cooling (1) **Additional guidance** Reference to cost must be qualified. E.g. comparable performance, upgrade costs etc	3

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Question Number	Answer	Mark
1b	Any **three** from: • Can include graphs/charts (to show progress). (1) • **Import** images (e.g. scans of work, photographs of the child). (1) • Formatting features (e.g. tables, bold etc. to structure reports). (1) • Mail merge – (to add details of specific child to standard letter/form). (1) • Able to print copies (to give to parents) (1) • Sharing/collaboration features (e.g. save to parent portal, user access, email to parents etc. (1) **Additional guidance** For Mark point 3 allow examples of formatting features For Mark point 6 allow examples of sharing/collaboration features Accept any feature of word-processing software that could be appropriate for producing learning records	3

Question Number	Answer	Mark
1c	Award **one** mark for the identification and **one** additional mark for the appropriate expansion to a maximum of **four** marks. • Compatibility issues (with parent's systems) (1) due to unusual/uncommon file formats (1) • May not have same features as proprietary software / Staff may be unsure how to perform a task (1) may need additional training / time to become familiar with software (1) • Updates may be infrequent (1) as some open-source projects are worked on part time/made by very small teams (1) • May be more difficult to access (official) help/support (1) due to smaller development team/user base /may have to rely on users/community (1) • Security vulnerabilities can be **spotted**/**exploited** (1) as original source code is available online (1)	4

Question Number	answer	**Mark**
1d	Example solution: Total $\begin{bmatrix} (26+29) & (34+19) \\ (42+42) & (27+41) \\ (21+32) & (26+18) \end{bmatrix}$ Total $\begin{bmatrix} 55 & 53 \\ 84 & 68 \\ 53 & 44 \end{bmatrix}$	
	Award: **one** mark for correct use of matrices in **calculations.** and **one** mark for correct final daily totals as a matrix	2

Question Number	**Answer**	**Mark**
1e	$\begin{bmatrix} 130 & 170 & 166 & 112 & 84 & 108 \end{bmatrix}$ **Additional guidance** Ignore use of brackets and delimiters.	1

Question Number	Answer	Mark
1f	Award **one** mark for the identification and **one** additional mark for the appropriate expansion to a maximum of **four** marks. • Array holds a single data type (1) attendance will always be an integer (1) • The array length is fixed (1) and there are always 6 days to be represented (1) • The index for each day/month will always be the same (e.g. Monday = [0], Tuesday [1] etc) (1) so code can be reused with new data each month / individual data points can be easily retrieved (1) • **Memory** efficient (1) because the size of the data set is known/predictable (1) • Needs to represent/store two groups of data (1) so can store day/month as **separate** dimensions (1)	4

Question Number	Answer	Mark
1g	Award **one** mark for the identification and **one** additional mark for **each** appropriate expansion to a maximum of **three** marks. • Ensures data is sensible/reasonable (1) by imposing rules on data entry (1) to ensure it is within specific limits/parameters/formats (1) • To reduce potential (runtime) errors (1) by alerting the user (1) if data does not meet requirements (1)	3

Total for Question 1 = 20 Marks

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Question Number	Answer	Mark
2a	Award **one** mark for each of: • Mobile broadband/4G/5G • **Public/Open** Wi-Fi hotspots • Home network/LAN (e.g. working from home, client's Wi-Fi connection) **Additional guidance** Allow "tethering", "phones data/internet" for Mark point 1 Do not award "Use the internet" (or similar) without specific reference to the connection method Allow "Wi-Fi" on its own once only Do not allow "Ethernet".	2

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Question Number	Answer	Mark
2b	Award **one** mark for the identification and **one** additional mark for the appropriate expansion to a maximum of **four** marks. • Don't need to employ specialist IT staff / nurses may not have technical skills (1) to maintain /manage a physical server (1) • Resources can be scaled (up/down) (1) to respond to changes in need (1) • Reduces the amount of physical devices in the office (e.g. server, racks, backups etc) (1) which reduces the office space required (1) • Cloud service provider could take care of upgrades/backups/security etc (1) allowing the charity staff to focus on their own jobs (1) • (Potentially) Improved data security / reduced chance of data loss (1) because data is off-site /protected from localised issues (e.g. flood, fire) (1) • Lower **initial** costs (1) such as the purchasing a server/setting up a server room (1)	4

Question Number	Answer	Mark
2c	Award **one** mark for the identification and **one** additional mark for the appropriate expansion to a maximum of **four** marks. • Firewall (1) to prevent unauthorised connections/access to devices and server (1) • Anti-malware (1) to prevent malicious software (1) • Encryption software (1) to make data unreadable during transmissions (1) • Network utilities (1) to monitor and log (suspicious) activity on their network/server (1) • VPN (1) to mask IP/secure network connections (1) • Password managers (1) to generate complex/unique passwords / to encrypt stored passwords (1) **Additional guidance** Allow brand names for Anti-malware/firewall where appropriate. Do not accept back-up	4

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Question Number	Answer	Mark
2d	A description such as: Removes (unnecessary) data (1) which reduces overall file size (1) so less bandwidth is used /less data to transfer (1)	3

Question Number	Answer	Mark
2e	Award **one** mark for each correct outcome. <table><tr><th>Username</th><th>Password</th><th>6 digit code</th><th>Security question</th><th>Access granted?</th></tr><tr><td>TRUE</td><td>TRUE</td><td>TRUE</td><td>FALSE</td><td>TRUE</td></tr><tr><td>TRUE</td><td>FALSE</td><td>FALSE</td><td>TRUE</td><td>FALSE</td></tr><tr><td>TRUE</td><td>TRUE</td><td>FALSE</td><td>TRUE</td><td>TRUE</td></tr></table> **Additional guidance** Allow any accepted representation of Boolean outcome (e.g. Yes/No, True/False 1/0 etc)	3

<table>
<tr><th>Question Number</th><th>Answer</th><th>Mark</th></tr>
<tr><td>2f</td><td>Award one mark for each correct output.
<table>
<tr><th>Test data</th><th>Final output</th></tr>
<tr><td>givenName = Mae
familyName = Calderon
yearStarted = 2024</td><td>MaeCald24</td></tr>
<tr><td>givenName = Jo
familyName = Mata
yearStarted = 2024</td><td>Error</td></tr>
<tr><td>givenName = Terry
familyName = Bowlesworth
yearSarted = 2023</td><td>TerBowl23</td></tr>
<tr><td>givenName = John
familyName = Poe
yearStarted = 2023</td><td>JohPoeP23</td></tr>
</table>
Additional guidance:
Ignore capitalisation</td><td>4</td></tr>
</table>

Total for Question 2 = 20 Marks

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Question Number	Answer	Mark
3a	A description to include **four** from: • Fewer bits are used to represent the image (1) • So fewer colors can be recreated (1) • Resulting in a less accurate/less precise image (1) • Which reduces the (overall) file size (1) • So less disk space is used / target game will be smaller (1) **Additional guidance** Allow "uses less data" for MKPT 4 Do not award reference to file type.	4

Question Number	Indicative content	Mark
3b	**A discussion of the factors Fran will need to consider when using emulation**. Create virtual versions of target platforms so Fran would not need physical versions of every phone, console etc. Emulation would allow her to tweak the specifications of the emulated platforms to identify optimum system requirements. Emulation is not 100% accurate so the results of testing in an emulated environment may not be reliable. Emulated platforms typically perform less effectively that the real thing as some performance/response is lost when converting between the emulation software and the host machine. Emulating some platforms may require licensing of operating systems etc from the manufacturers which may increase development costs. Fran may need to seek permission to emulate some platforms to avoid copyright infringement, which could increase development time while permission is sought.	6

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Level	Mark	
	0	No rewardable material
1	1-2	Demonstrates isolated elements of knowledge and understanding, there will be major gaps or omissions Few of the points made will be relevant to the context in the question Limited discussion which contains generic assertions rather than considering different aspects and the relationship between them
2	3-4	Demonstrates some accurate knowledge and understanding, with only minor gaps or omissions Some of the points made will be relevant to the context in the question, but the link will not always be clear Displays a partially developed discussion which considers some different aspects and some consideration of how they interrelate, but not always in a sustained way
3	5-6	Demonstrates mostly accurate and detailed knowledge and understanding Most of the points made will be relevant to the context in the question, and there will be clear links Displays a well-developed and logical discussion which clearly considers a range of different aspects and considers how they interrelate, in a sustained way

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<table>
<tr><th>Question Number</th><th>Indicative content</th><th>Mark</th></tr>
<tr><td>3c</td><td>An analysis of how the operating system will manage system components and tasks when executing the game.

Networking
Ensure that the computer communicates correctly with the network access point.
Ensures data sent/received is processed using the correct protocol up to the transport layer.
Would manage the device driver for the computers network interface card to ensure that the game and the network card can communicate.

Multi-tasking
Monitors tasks and processes and CPU allocation to ensure that the computer’s core functions can function as well as ensuring there is sufficient CPU time for the game.
Uses time slicing to allocate instructions to a single processor on an ‘as needed’ basis and manages the allocation of processes to different processors in a multi core/multiprocessor system.

Interrupts
Player input such as keystroke/gamepad to control an in game character
Inter-processor interrupt – if the system uses more than one processor one processor would interrupt the other (e.g. if instructions are being processed by different processors the most vital instruction will take over)

System critical information/monitoring information (e.g. Processor overheating, virus alerts etc)
In game processes that are flagged as more important such as:
Player pressing ‘pause’
Instructions relating to ‘in game’ action would be of higher importance than background action
Player to player messages
Memory management
Allocates and deallocates the memory locations for instructions and data.
Monitors levels of data in memory and frees up memory as required ready for use by other instructions.
Manages the fetch-decode-execute cycle</td><td>8</td></tr>
</table>

	Monitors Physical memory and utilises virtual memory as required to help maintain system performance.	

	0	No rewardable material
1	1-3	Demonstrates isolated knowledge and understanding, there will be major gaps or omissions Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question Limited analysis which contains generic assertions rather than interrelationships or linkages
2	4-6	Demonstrates some accurate knowledge and understanding, with few minor omissions/any gaps or omissions are minor Breaks the situation down into component parts and some of the points made will be relevant to the context in the question Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
3	7-8	Demonstrates mostly accurate and thorough/detailed knowledge and understanding Breaks the situation down into component parts and most of the points made will be relevant to the context in the question Displays a well-developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner

Total for Question 3 = 18 Marks

Question Number	Indicative content	Mark
4a	A discussion of ways that the company could use computer hardware to protect its data and computer systems. Discussion could include reference to: • HW based firewalls. • Biometric devices/fingerprint scanners on computers to restrict access. • Card readers to prevent access to key areas of the factory. • Device hardening techniques. • **Removable** drives that are taken off site/moved to secure location to protect data backups Refences to the scenario may include: • Importance of protecting personal data to ensure they do not breach DP laws. • Sensitivity of business-related data and ensuring it is not accessible to those that should not see it including internal workers and rival companies. • Trade-off between ensuring production is not disrupted/data alerting staff to potential issues which could cause problems to the business and keeping sensitive data safe. • Quite a large company, but only certain staff will need access to the data. E.g. most staff on the production line will not need any of the data. • Considering processes for collecting and managing biometric data, who will need to provide it (only have to collect the biometric data of those that need access to the systems) • Impact and issues relating to collecting biometric data vs issuing, and managing access cards.	10

Level	Mark	
	0	No rewardable material
1	1-4	Demonstrates isolated elements of knowledge and understanding, there will be major gaps or omissions Few of the points made will be relevant to the context in the question Limited discussion which contains generic assertions rather than considering different aspects and the relationship between them
2	5-7	Demonstrates some accurate knowledge and understanding, with only minor gaps or omissions Some of the points made will be relevant to the context in the question, but the link will not always be clear Displays a partially developed discussion which considers some different aspects and some consideration of how they interrelate, but not always in a sustained way
3	8-10	Demonstrates mostly accurate and detailed knowledge and understanding Most of the points made will be relevant to the context in the question, and there will be clear links Displays a well-developed and logical discussion which clearly considers a range of different aspects and considers how they interrelate, in a sustained way

Question Number	Indicative content	Mark

4b	An evaluation of the suitability of the company's current backup and storage systems.
	
	Points that may be considered:
	• Relative performance of SSD and HDD including:
	    o read/write speeds
	    o durability
	    o physical wear and tear of plates (HDD) vs limited number of writes to a specific location (SDD)
	• Use cases e.g. SDD to store programs/OS and HDD to store data
	• Potential use of RAID arrays
	• Cost and shelf life of external media for backup vs RAID
	• Need to ensure external backups are kept safe/off site
	• Remembering to initiate backup every night and issues that may occur if this isn't done.
	
	Relation to scenario
	• Some data is quite static (e.g. staff personal data) whereas business data and production data may change rapidly.
	• Need to ensure production data can be restored quickly, so in the event of an issue business can resume before significant losses are made – also need to ensure that any data generated that day (up to the point of failure) is somehow backed up and restored as over production or errors in orders could result in loss of profits.
	
	The learners should make some supported judgments of which specific devices they would suggest to meet the needs of the company.
		12

Level	Mark	
	0	No rewardable material
1	1-4	Technical vocabulary is used but is not used appropriately to support arguments in relation to the issues of the question. Few of the points made will be relevant to the context in the question. Limited evaluation which contains generic assertions leading to a conclusion (if present) that is superficial or unsupported
2	5-8	Accurate technical vocabulary is used to support arguments but not all are relevant to the issues of the question Some of the points made will be relevant to the context in the question, but the link will not always be clear. Displays a partially developed evaluation which considers some different competing points, although not always in detail, leading to a conclusion which is partially supported.
3	9-12	Fluent and accurate technical vocabulary is used to support arguments that are relevant to the issues of the question Most of the points made will be relevant to the context in the question, and there will be clear links Displays a well-developed and logical evaluation which clearly considers different aspects and competing points in detail, leading to a conclusion that is fully supported.

Total for Question 4 = 22 Marks

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